



ACADEMIC REGULATIONS & SYLLABUS

Faculty of Computer Science & Applications
Smt. Chandaben Mohanbhai Patel Institute of
Computer Applications
MCA Lateral Programme





CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Computer Science & Applications

ACADEMIC REGULATIONS

Master of Computer Application (Lateral) Programme
(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
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www.charusat.ac.in

Year 2018-19

CHARUSAT

ACADEMIC RULES

Master of Computer Applications (MCA) Lateral Programme

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Duration of Programme

- | | |
|-----|---|
| (i) | Postgraduate programme : Master of Computer Applications |
| | Minimum 4 semesters (2 academic years) |
| | Maximum 6 semesters (3 academic years) |

3. Eligibility & Mode of admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

4. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

5. Attendance

- 5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from

attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

5.2 Student attendance in a course should be 80%.

6 Course Evaluation

6.1 The performance of every student in each course will be evaluated as follows:

- 6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
- 6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

6.2 University Examination

- 6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.
- 6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

6.3 Performance at Internal & University Examination will be done on the relative grading system.

7 Grading

7.1 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

(i) $SGPA = \sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester

(ii) $CGPA = \sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

- (iii) Students having CGPA of at least 3.0 will be allowed to go to 3rd semester (i.e. Second year)
- (iv) A) Students having backlogs of 2 or less together in 1st and 2nd semester (i.e. First year) and /or backlogs of 4 or less together in 3rd and 4th semesters (i.e. Second year)

And

 B) Students having CGPA of at least 4.0 will be allowed to go to 5th semester (i.e. Third year)
- (v) In addition to above, the student has to complete the required formalities as per the regulatory bodies.

8 Awards of Degree

8.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

- 8.1.1 He should have earned at least minimum required credits as prescribed in course structure; and
- 8.1.2 He should have cleared all evaluation components in every course; and

8.1 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

9 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA ≥ 7.5 & ≤ 10
First class:	CGPA ≥ 6.0 & < 7.5
Second Class:	CGPA ≥ 5.0 & < 6.0
Pass Class:	CGPA < 5.0

10 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)

FACULTY OF COMPUTER SCIENCE & APPLICATIONS

CHOICE BASED CREDIT SYSTEM

FOR

MASTER OF COMPUTER APPLICATION (LATERAL)

CHOICE BASED CREDIT SYSTEM

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses:

Foundation Courses, Core Courses and Elective Courses.

Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

Vision

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

Program Educational Objectives (PEOs)

Graduates of Master of Computer Applications(MCA) program at the Charotar University of Science and Technology (CHARUSAT) should be able to utilize the knowledge gained from their academic program to :

- PEO1 Be successful professionals in industry, government, academia, research, entrepreneurial pursuit and consulting firms.
- PEO2 Integrate fundamentals and advanced concepts of computer science and applications to develop innovative solutions to societal problems.
- PEO3 Function and communicate effectively, both individually and within multidisciplinary teams.
- PEO4 Pursue life-long learning to fulfill their goals.
- PEO5 Be sensitive to the consequences of their work, both ethically and professionally, for productive professional careers.

Program Outcomes (POs)

The curriculum of Master of Computer Applications(MCA) program at the Charotar University of Science and Technology (CHARUSAT) is designed to assure students, at the time of graduation are able to:

- PO1 Identify, critically analyze, formulate and develop computer applications.
- PO2 Design a computing system to meet desired needs within realistic constraint.
- PO3 Use modern computing tools and techniques for effective software engineering.
- PO4 Devise and conduct experiments, interpret data and provide well informed conclusions.
- PO5 Have a fundamental capability in oral and written communication.
- PO6 Function effectively within multidisciplinary teams.
- PO7 Understand the impact of system solutions in a contemporary and societal context for sustainable development.
- PO8 Function professionally with ethical responsibility.
- PO9 Appreciate the importance of goal setting and to recognize the need for life-long learning.

Teaching and Examination Scheme (MCA Lateral Programme)
Effective from Year 2018-19

Semester-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA918-921	Elective-III	4	3	7	7	10	20	70	15	15	70	200
CA922	Open Source Frameworks	4	3	7	7	10	20	70	15	15	70	200
CA923	Data Mining and Analytics	4	3	7	7	10	20	70	15	15	70	200
CA924	Mini Project	-	6	6	6	-	-	-	30		70	100
		12	15	27	27	300			400			700

Elective-III

1. CA918- Collaboration Technologies (ONLY for students who have completed Windows Communication Foundation Framework offered in Elective Set I)
2. CA919-Web Technologies (ONLY for students who have completed Frameworks and Applications offered in Elective Set I)
3. CA920-Advanced Management Information System
4. CA921-Intrusion Detection and Computer Forensics

Teaching and Examination Scheme (MCA Lateral Programme)
Effective from Year 2018-19

Semester-IV

Course Code	Course Title	Teaching Scheme				Internal	End Semester Examination		Total
		Contact Hours			Credit	Continuous Evaluation	Report	Presentation & Viva	
		Inst.	Industry	Total					
CA925	Project Work	2	28	30	30	200	200	400	800

OBJECTIVES, TEACHING SCHEME & DETAILED SYLLABUS

FOR

**MCA Lateral PROGRAMME
(3rd SEMESTER)**

**EFFECTIVE FROM
ACADEMIC YEAR 2018-19**

Semester-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA918-921	Elective-III	4	3	7	7	10	20	70	15	15	70	200
CA922	Open Source Frameworks	4	3	7	7	10	20	70	15	15	70	200
CA923	Data Mining and Analytics	4	3	7	7	10	20	70	15	15	70	200
CA924	Mini Project	-	6	6	6	-	-	-	30		70	100
		12	15	27	27	300			400			700

Elective-III

1. CA918- Collaboration Technologies (ONLY for students who have completed Windows Communication Foundation Framework offered in Elective Set I)
2. CA919-Web Technologies (ONLY for students who have completed Frameworks and Applications offered in Elective Set I)
3. CA920-Advanced Management Information System
4. CA921-Intrusion Detection and Computer Forensics

CA918 - Collaboration Technologies

(200 Marks)

Contact Hours: 07

Prerequisites: CA835: Windows Communication Foundation Framework.

Methodology & Pedagogy:

During theory lectures illustrations emphasizing the need for various features of collaboration technologies will be given. During practical sessions, students will be required to develop intranet applications using concepts discussed during class. Students will be required to submit a report on result of group task assigned to them at beginning of course.

Learning Outcomes: Upon successful completion of the syllabus students shall have working understanding of collaborative technologies available in industry. Students will also be able to develop, test and deploy intranet applications independently using collaborative technologies

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Introduction to collaboration technologies	08	36
2	Introduction to SharePoint	09	
3	Working with List & Library	09	
4	Data manipulation and business processes	08	
5	SharePoint Branding and Personalization	08	
6	Security with SharePoint	06	
	Total	48	36

Detailed Syllabus:

Unit I: Introduction to collaboration technologies Hours: 08

Introduction to collaboration technologies, Evolution of Collaboration Technology, Comparison between different collaboration technologies, Collaborative work systems, Level of Collaboration, Collaborative management tools. Collaborative software and human interaction.

Unit II: Introduction to SharePoint Hours: 09

Overview of SharePoint Architecture : Server Object Model, Objectives, features and components of SharePoint, Role of SharePoint Server in intranet, Versions history of SharePoint, SharePoint server Editions, SharePoint Installation and Administration, Developing and sharing sites using SharePoint: Web application, Site Collection and Sites. SharePoint Services, searching information with SharePoint.

Unit III: Working with List & Library Hours: 09

Overview of List, Type of List: Library, Communication, Tracking and Custom, Creating, Adding and Editing List Items in List, Managing document with document library, Managing media using asset library

Unit IV: Data manipulation and business processes Hours: 08

Configuring data source, Manipulating data with ASP.NET data controls, Overview of XSLT list view and data view. Working with data view and data form web parts, Overview with custom workflow and custom list actions.

Unit V: SharePoint Branding and Personalization Hours: 08

Introduction to SharePoint Branding: Branding with themes, CSS , webparts and master pages, Creating and managing blogs: Managing blog posts, Managing blog comments, Creating and managing personal sites: Editing user profiles, Uploading content, Using note board,

Unit VI : Security with SharePoint Hours: 06

Introduction to security in SharePoint, Managing SharePoint groups, Creating and granting permissions to groups, Granting access to Lists, Libraries and individual items, Introduction to claim based authentication.

Core Books:

1. Kathy Hughe; SharePoint Designers 2010, Unleashed, SAMs publication
2. Olga M. Londer, Penelope Coventry; Microsoft SharePoint 2013 Step by Step, Microsoft Press
3. Jonathan Lightfoot and Chris Beckett; Microsoft SharePoint 2010: Plain & Simple, Microsoft Press

Reference Books:

1. Ted Pattison ; Andrew Connell ; Scot Hillier, Inside Microsoft SharePoint 2010, Microsoft Press
2. Paolo Pialorsi: Microsoft SharePoint 2010: Developer's reference, Microsoft Press

Web Reference:

1. www.sharepoint-tutorial.net [For SharePoint Installation]
2. www.wonderlaura.com [Blog on SharePoint]

Course Outcomes: Upon successful completion of the course, students will be able to:

C01 : Understand the basics of collaboration technologies.

C02 : Learn the tool share point for collaboration in software project.

C03 : Understand branding and personalization in share point.

CO4 : Understand security aspects in share point.

3. Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	C02	C03	C04
1	Introduction to collaboration technologies	✓			
2	Introduction to SharePoint	✓	✓	✓	✓
3	Working with List & Library	✓	✓	✓	✓
4	Data manipulation and business processes	✓	✓	✓	✓
5	SharePoint Branding and Personalization			✓	
6	Security with SharePoint				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	1	2	2	-	-	3	2	1	2
CO2	3	2	2	-	-	3	2	1	2
CO3	3	2	2	-	-	3	2	1	2
CO4	3	2	2	-	-	3	2	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA919: Web Technologies

(200 Marks)

Contact Hours: 07

Pre-requisite: CA831: Enterprise Computing, CA836: Frameworks and Applications.

Methodology & Pedagogy: During theory sessions the students shall be introduced to common techniques and technologies for developing web based application and they shall explore Struts framework, AJAX and Web Services. During practical sessions students will be trained to develop various web applications using the studied frameworks.

Learning Outcomes: Upon successful completion of the syllabus students shall be able to understand the concepts of web technologies and apply web technologies concepts to develop web applications.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Ajax Framework	06	36
2	The Struts Framework : Basics	08	
3	The Struts Framework: Views, Results and Interceptors	10	
4	The Struts Framework: Validation & Advanced Topics	10	
5	Web Services Fundamentals	06	
6	Java Web Services	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit - I: AJAX Framework Hours: 06

Ajax introduction, working, technologies, working with Ajax XMLHttpRequest – XHR Create Object, XHR Request, XHR Response

Unit - II: The Struts Framework: Basics Hours: 08

Introduction - Brief History of the Web, Framework, Creation of the Struts Framework, Struts1 and Struts2 Architecture, Configuration Files, Filter Dispatcher, Actions, Interceptors, Results and Result Types, Working with Actions

Unit - III: The Struts Framework: Views, Results and Interceptors Hours: 10

Building the views – Struts Tags, Data Tags, Control Tags, Other JSTL Tags

Results – Dispatching requests and responses, commonly used result types

Using Interceptors – Built-in interceptors, pre-configured stacks, mapping interceptors, customized interceptors

Unit - IV: The Struts Framework: Validation & Advanced Topics Hours: 10

Validator framework – architecture, workflow, wiring actions, custom validator, Resource Bundles

Advanced topics – Annotations, Alternative method invocation, Dynamic Method invocation, Displaying wait pages automatically, CRUD operations, Tiles plugin

Unit - V: Web Services Fundamentals Hours: 06

Overview of traditional distributed computing and its drawbacks, Web Service overview and its advantages over distributed computing

Service Oriented Architecture (SOA) with its benefits, SOAP and Service Description overview. Interpretability: Web Service protocol support, WSE.

Unit - VI: Java Web Services Hours: 08

Types of Web Services: SOAP based services and RESTful services and difference between them, JAVA's SOAP API, All about WSDL, SOAP handling, Implementing SOAP based Web Service

RESTful services: What is Rest? From @WebService to @WebServiceProvider, Implementing RESTful Web service as HTTPServlets.

Core Books:

1. Sharanam Shah, Vaishali Shah: Struts 2 for Beginners, Shroff Publishers & Distributors, 2nd Edition
2. Donald Brown, Chad Michael Davis, Scott Stanlick : Struts 2 in Action, Dreamtech press publication
3. Martin Kalin: Java Web Services Up and Running by, O'Reilly Publication

Reference Books:

1. Chuck Cavaness: Programming Jakarta Struts, O'Reilly Publication
2. Ethan Cerami: Web Services Essentials, O'Reilly Publication

Web References:

1. http://www.tutorialspoint.com/struts_2/index.htm [For struts examples]
2. <http://www.w3schools.com/ajax/default.asp> [For ajax examples]
3. <http://javapapers.com/category/web-service/> [For web service examples]
4. <http://www.ws-i.org/> [For web service examples]

Course Outcomes: Upon successful completion of the course Students will be able to:

CO1 : Understand different web technology frameworks

CO2 : Develop real world applications using them.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes	
		CO1	CO2
1	Ajax Framework	√	√
2	The Struts Framework : Basics	√	√
3	The Struts Framework: Views, Results and Interceptors	√	√
4	The Struts Framework: Validation & Advanced Topics	√	√
5	Web Services Fundamentals	√	√
6	Java Web Services	√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	3	3	2	-	-	2	-	2
CO2	3	3	3	2	2	3	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA920: Advanced Management Information System

(200 Marks)

Contact Hours: 07

Pre-requisite: None.

Methodology & Pedagogy: During the Theory sessions, basic concepts of ERP will be covered with integration of CRM and SCM. Emphasis will be given on how Business Intelligence can have impact on organization success. Industrial visits will be arranged in order to demonstrate actual functioning of such systems. Students will be required to carry out a case study of such system.

Learning Outcomes: Student will be able to understand enterprise resource planning as a management approach and how information systems can help promote ERP, understanding of the components of ERP systems and the differences between them. Also they will learn about business value of integrating supply chain management, customer relationship management, and enterprise resource planning systems together, how an organization can measure ERP success, choose ERP software, and use ERP in markets.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to ERP	07	36
2	ERP Business Modules and ERP Market	08	
3	ERP Implementation	09	
4	ERP Present and Future	07	
5	Customer Relationship Management and Supply Chain Management	08	
6	Business Analytics	09	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:**Unit - I: Introduction to ERP Hours: 07**

Introduction to ERP, Basic ERP Concepts, Justifying ERP Investments, Risks of ERP, Benefits of ERP.

Unit - II: ERP Business Modules and ERP Market Hours: 08

Introduction to Business Modules of an ERP Package, ERP Marketplace and Dynamics, SAP AG, Oracle Corporation, PeopleSoft.

Unit - III: ERP Implementation Hours: 09

Implementation challenges, ERP Implementation Strategies, ERP Implementation Life Cycle, Pre Implementation Tasks, Requirements Definition, Implementation Methodologies.

Unit - IV: ERP Present and Future Hours: 07

Enterprise Application Integration (EAI), ERP and e-business, ERP, Internet and WWW – ERP II, ERP and Total Quality Management, Future Directions and Trends in ERP.

Unit - V: Customer Relationship Management and Supply Chain Management Hours: 08

Introduction to CRM, Introduction to eCRM, Difference between CRM & eCRM, Features of eCRM, Implementing CRM, Introduction to Supply Chain Management, Creating Outcome-Driven Tasks and Processes, Transportation Systems.

Unit - VI: Business Analytics Hours: 09

ERP and Related Technology, Business Intelligence, Business Process Reengineering, Data Warehousing, Data Mining, On-Line Analytical Processing (OLAP).

Core Books:

1. Alexis Leon: ERP Demystified, Tata Mc Graw Hill, 2008.
2. Thomas F. Wallace, Michael H. Kremzar: ERP: Making It Happen: The Implementers' Guide to Success with Enterprise Resource Planning, John Wiley & Sons, Inc., 2001.

Reference Books:

1. K. C. Laudon, J. P. Laudon: Management Information Systems, 11th Edition, Pearson Education, 2010.
2. W.S. Jawadekar: Management Information Systems, Tata Mc Graw Hill, 2004.

Web References:

1. http://www.tutorialspoint.com/management_concepts/enterprise_resource_planning.htm
[For ERP Concepts]
2. www.lesman.com/downloads/CRM2011-Step-by-Step.pdf [For CRM Implementation]

Course Outcomes: Upon successful completion of the course, students will be able to:

C01 : Understand fundamentals and different modules of ERP.

C02 : Learn Supply chain Management and Customer Relationship Management effectively.

C03 : Understand different business analytics.

1. Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes		
		C01	C02	C03
1	Introduction to ERP	✓		
2	ERP Business Modules and ERP Market	✓		
3	ERP Implementation	✓		
4	ERP Present and Future	✓		
5	Customer Relationship Management and Supply Chain Management		✓	
6	Business Analytics			✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	2	2	-	-	-	2	-	2
CO2	3	2	2	-	-	-	2	-	2
CO3	3	2	2	-	-	-	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA921: Intrusion Detection and Computer Forensics

(200 Marks)

Contact Hours: 07

Pre-requisite: Operating Systems concepts and Network Technology.

Methodology & Pedagogy: During theory lectures, different types of network attacks will be discussed. Also, detection and prevention of attacks will be discussed. Through case study based approach students will be asked to identify different attacks and how to resolve it.

Learning Outcomes: Upon successful completion of this course, the students should be able to: Describe different threats and attacks to networks, How to identify them and solving out the problems related to it. Student will also get idea about the Fundamental of Computer Forensics & how it will be useful in real life.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Concepts and Definitions	07	36
2	System Security	08	
3	Network Attacks	08	
4	Analysis Schemes	10	
5	Responses	07	
6	Computer Forensics Fundamentals	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit - I: Concepts and Definitions Hours: 07

An introduction to Intrusion Detection, Security Concepts: Definition of Computer Security, Threat, Vulnerability, Security Policy

Intrusion Detection Concepts: Architecture, Monitoring Strategy, Analysis Type, Timing, Goals of Detection, Control Issues, Determining Strategies for Intrusion Detection,

Network defense tools: Firewalls, Intrusion Detection, Traffic shapers, and filters

Unit - II: System Security Hours: 08

Intruders, Intrusion detection, Password management

Malicious software: Viruses and related threats, virus countermeasures

Firewalls: Firewall design principles – Firewall configurations – Trusted systems

Unit - III: Network Attacks Hours: 08

Attack Taxonomies, Privilege Escalation Attack: Buffer Overflow Attacks, Race condition, Man-in-the-middle

Denial of Service (Dos) and Distributed Denial of Service (DDoS) Attacks: Detection approaches for Dos & DDoS Attacks, Prevention and Response for DoS & DDoS Attacks, Examples related to it.

Worm Attacks: Modeling and Analysis of Worm Behaviors, Detection and Monitoring, Worms Containment, Examples

Routing Attacks: OSPF Attacks, BGP Attack

Unit - IV: Analysis Schemes Hours: 10

Defining Analysis, Detecting Intrusion Model for Intrusion analysis: Performing Analysis, Feedback and refinement

Techniques:

Misuse Detection: Pattern Matching, Rule-based Techniques, State-based Techniques

Anomaly Detection: Advanced Statistical Models, Rule based Techniques, Biological Models

Alternative detection Schemes, Specification Based Detection, Hybrid Detection

Unit - V: Responses Hours: 07

Requirements for response: Operational Environment, System purpose & priorities, Regulatory or statutory requirements

Types of Response: Active Response, Passive Response

Unit - VI: Computer Forensics Fundamentals Hours: 08

Introduction to Computer Forensics, Use of Computer Forensics in Law, Computer Forensics Services

Computer Forensics Technology: Type of Military Computer Forensics Technology, Type of Law Enforcement Computer Forensics Technology, Type of Business Computer Forensics Technology, Specialized Forensics Techniques, Hidden Data and how to find it, Spyware and adware, Internet Tracing Methods, Security and wireless Technologies

Systems: Internet Security Systems, Intrusion Detection Systems, Firewall Security Systems

Core Books:

1. Rebecca Gurley Bace: Intrusion detection, ISBN: 1-57870-185-6, Macmillan Technical Publishing.
2. Ali A. Ghorbani, Wei Lu, Mahbod Tavallaee: Network Intrusion Detection and Prevention, Springer, ISBN : 978-0-387-88771-5.
3. Paul E. Proctor : The Practical Intrusion Detection Handbook, Pearson Publication, ISBN-10: 0130259608 , ISBN-13: 9780130259608.
4. John R. Vacca : Computer Forensics Computer Crime Scene Investigation, Volume 1, 2nd Edition, Course Technology.
5. Ryan Trost : Practical Intrusion Analysis Prevention and detection for the Twenty – First Century, Pearson Publication, ISBN-13 : 978-0-321-59180-7.

Reference Books:

1. Michael Sheetz: Computer Forensics: An essential guide for accountants, lawyers, and managers, John Wiley & Sons, ISBN: 978-0-471-78932-1.
2. Stephen Northcutt, Judy Novak: Network Intrusion Detection, 3rd Edition, New Riders Publishing.

Web References:

1. http://www.routeralley.com/ra/docs/intro_ids.pdf [For Intrusion Detection]
2. <http://opensecuritytraining.info/HTID.html> [For Intrusion Detection]
3. http://en.wikipedia.org/wiki/Computer_forensics [For Computer Forensic]

Course Outcomes: Upon successful completion of the course, students will be able to:

- C01 : Understand the fundamentals of Intrusion Detection.
- C02 : Learn about different categories of network attacks.
- C03 : Learn about analysis schemes in order to prevent attacks.
- C04 : Understand fundamentals of computer forensics.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	C02	C03	C04
1	Concepts and Definitions	✓			
2	System Security	✓			
3	Network Attacks		✓		

4	Analysis Schemes			✓	
5	Responses			✓	
6	Computer Forensics Fundamentals				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	1	2	1	-	-	-	2	2	2
CO2	2	2	1	-	-	-	2	2	2
CO3	2	2	1	-	-	-	2	2	2
CO4	2	2	1	-	-	-	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA922: Open Source Frameworks (200 Marks)

Contact Hours: 07

Prerequisite: CA839: Open Source Technology.

Methodology & Pedagogy: During theory lectures the emphasis will be given on the Development of WordPress based web application; students will be introduced to framework standards like theme and plug-in development.

Learning Outcomes: Upon successful completion of the syllabus students shall be able to develop Blog and CMS web application using WordPress. During Practical sessions, students will implement the concepts which are discussed in lecture.

Outline of the Course:

Sr.No.	Title of the Unit	Minimum No of Hours	
		Theory	Practical
1.	Introduction to WordPress	06	36
2.	Content management & Creating Widget in wordPress	08	
3.	Basic Theming in WordPress	09	
4.	Advance Theme in WordPress	09	
5.	Plugin Creation in WordPress	08	
6.	Users and Roles & Performance and Security	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus :

Unit – I: Introduction to WordPress Hours: 06

Introduction to framework, Why wordpress and its benefits, Sites built with wordpress , Installing and upgrading wordpress, installing themes, installing plugins, working with Dashboard and Settings.

Unit- II: Content management & Creating Widget in WordPress Hours: 08

Working with content, Post, Pages, Media Files, links and feeds, importing blogs and HTML files, after importing fixing what's broken. Creating Widget: Basic Widget, Dashboard Widget

Unit – III: Basic Theming in WordPress Hours: 09

Creating a Basic Theme: Using Widgets, Using Menus, Using Header and Background images, Turning HTML into a Theme, Template Tags, Theme Files, Theme File hierarchy, Listing Comments, Including Additional Files, Styling Content, Changing Appearance Based on Context, Creating Navigation Menus, Theme Functions, Child Themes.

Unit – IV: Advance Theme in WordPress Hours: 09

Modifying the Loop, Multiple Loops, Accessing Post Information outside the loop search, Engine Optimization, Adding Scripts and Style sheets, Database Errors and Maintenance Messages, Theme Options, Theme Frameworks, Distributing Themes.

Unit – V: Plugin Creation in WordPress Hours: 08

Creating a Template Tag, Adding an Options Page, Settings API, Wrapping Up, Plugin Possibilities, and Hooks: Filters and Actions, Variations on Option Pages, Publishing a Plugin.

Unit – VI: Users and Roles & Performance and Security Hours: 08

Users and Roles: Users, Extending User Profiles, Roles, And Roles in Action: Notifications, Moderation and workflow, Changing, Creating and Removing Roles.

Performance and Security : Caching, Permalinks and Performance, Tracking Down Performance Problems, Security Logins, Removing The Meta Generator Tag, File Permissions, Changing File Locations, Database Security

Core Books:

1. Stephanie Leary: Beginning WordPress 3, Apress Publication.
2. Ric Shreves and Jean-Baptiste Jung: WordPress 3 CookBook, PACKT publishing
3. Rachel McCollin, Tessa Blakeley Silver: WordPress Theme Development, 3rd Edition, PACKT publishing

Reference Books:

1. Jeff Siarto : Head First Wordpress, O'Reilly Publication
2. Brian Bondari, Everett Griffiths : WordPress 3 Plugin Development Essentials, PACKT publishing
3. Kent Mauresmo and Anastasiya Petrova : SEO for WordPress: How To Get Your Website on Page #1 of Google...Fast! (Volume 1), Vallon Investments, Inc Publication

Web References:

1. http://codex.wordpress.org/WordPress_Lessons [For Basic and Advance concept of Wordpress]
2. <http://www.siteground.com/tutorials/wordpress/> [Basic and Advance concept with video lectures]
3. <http://videos.wpbeginner.com/> [wordpress video tutorial]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Understand the basics of WordPress in web application development.
C02 : Be able to work with Widgets.
C03 : Be able to work with Themes in WordPress.
C04 : Understand Plugin integration and creation in the web application development.
C05 : Be able to develop the Role based Secure web applications.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to WordPress	✓		✓		✓
2	Content management & Creating Widget in WordPress		✓	✓		
3	Basic Theming in WordPress		✓	✓	✓	
4	Advance Theme in WordPress			✓	✓	
5	Plugin Creation in WordPress			✓	✓	
6	Users and Roles & Performance and Security			✓		✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
C01	2	-	-	2	-	-	3	2	2
C02	2	2	-	-	-	-	3	-	-
C03	-	3	2	-	-	2	-	-	-
C04	3	3	2	-	2	-	2	2	2
C05	2	3	-	2	3	3	2	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA923: Data Mining and Analytics

(200 Marks)

Contact Hours: 07

Pre-requisite: CA834: Advanced Database Technologies.

Methodology & Pedagogy: During theory lectures concepts of data mining with related technologies will be discussed. In practical sessions data mining tools will be introduced and will be used to design meaningful applications. Through case study based approach how data mining concepts are to be implemented in real application will be discussed.

Learning Outcomes: Upon successful completion of the syllabus students shall be able to learn how to develop data warehouse of any organization and how to retrieve hidden and useful data from large data set using techniques of data mining.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Data Warehousing & Mining	09	
2	Data Warehouse and OLAP Technology for Data mining	06	
3	Data Pre-processing and Data Mining Primitives	08	
4	Mining Association rule in large Databases	09	
5	Classification and cluster analysis	09	
6	Applications and Trends in Data Mining	07	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit - I: Introduction to Data Warehousing & Mining Hours: 09

An overview and definition along with clear understanding of the four key-words appearing in the definition. Differences between Operational Database Systems and Data Warehouses; Difference between OLTP & OLAP. Overview of Multi-dimensional Data Model, and the basic differentiation between “Fact” and “Dimension”; Multi-dimensional Cube. Concept Hierarchies of “Dimensions” Parameters: Examples and the advantages. Star, Snowflakes, and Fact Constellations Schemas for Multi-dimensional Databases. Measures: Their Categorization and Computation.

An Overview: What is Data Mining, Data Mining – on What Kind of Data, Data Mining

Functionalities – What Kind of Patterns Can be Mined, Concept/Class. Description: Characterization & Discrimination; Mining Frequent Patterns, Associations, and Correlations, Classification & Prediction; Cluster Analysis, Outlier Analysis. Classification of Data Mining Systems, Data Mining Task Primitives, Integration of a Data Mining System with a Database or Data Warehouse System, Major Issues in Data Mining.

Unit - II: Data Warehouse and OLAP Technology for Data mining Hours: 06

The Design of a Data Warehouse: A Business Analysis Framework; The Process of Data Warehouse Design, A 3-Tier Data Warehouse Architecture; Enterprise Warehouse, Data mart, Virtual Warehouse.

Unit - III: Data Pre-processing and Data Mining Primitives Hours: 08

The need for Pre-processing, Descriptive Data Summarization, Data Cleaning: Missing Values, Noisy Data, Inconsistent Data, Data Integration & Transformation. Data Cube Aggregation; Attribute Subset Selection, Dimensionality Reduction: Basic Concepts with example. Reduction Methods: Regression & Log-linear Models, Histograms, Clustering, Sampling. Introduction to Data Discretization & Concept Hierarchy Generation.

Unit - IV: Mining Association rule in large Databases Hours: 09

Basic Concepts: Market Basket Analysis; Frequent Itemsets, Closed Itemsets, and Association Rules. Frequent Pattern Mining: A Roadmap Apriori Algorithm: Finding Frequent Item sets Using Candidate Generation; Generating Association Rules from Frequent Item sets; Improving the Efficiency of Apriori. From Association Mining to Correlation Analysis; Strong Rules Are Not Necessarily Interesting: An Example; From Association Analysis to Correlation Analysis.

Unit - V: Classification and cluster analysis Hours: 09

An Overview, Basics of Supervised & Unsupervised Learning, Difference between Classification & Prediction, Criteria for Comparing Classification. Methods, Classification by Decision Tree Induction, Tree Pruning, Extracting Classification Rules from Decision Trees. Bayesian Classification: Bayes Theorem, Naïve Bayesian Classification, Bayesian Belief Networks, An Overview of Other Classification Methods. Prediction: Linear & Multiple Regression, Non-linear Regression, Other Regression Models. Classifier Accuracy: Estimating the Accuracy, Increasing the Accuracy, Accuracy Parameters – Sensitivity, Specificity, Precision.

Unit - VI: Applications and Trends in Data Mining Applications Hours: 07

Data mining research potential. Selection of data mining tool and technique. Example of

commercial data mining applications.

Application of data mining in real word problems like banking sector, telecommunication, mobile application.

Another aspect of data mining: Text Mining, Web Mining, Education Data Mining and Opinion Mining.

Core Books:

1. Jiawei Han & Micheline Kamber: Data Mining: Concepts & Techniques, 2nd Edition, Morgan Kaufmann Publishers.

Reference Books:

1. Paulraj Ponniah: Data Warehousing Fundamentals: A Comprehensive Guide for IT Professionals, Wiley-India.
2. Daniel T. Larose: Data Mining Methods & Models, Wiley-India.
3. Vikram Pudi & P. Radhakrishnan: Data Mining, Oxford University Press.
4. Alex Berson & Stephen J. Smith: Data Warehousing, Data Mining & OLAP, Tata McGraw-Hill.
5. Michael J. A. Berry & Gordon S. Linoff: Data Mining Techniques, Wiley-India.
6. Pang-Ning Tan, Michael Steinbach, Vipin Kumar: Introduction to Data Mining, Pearson Education.
7. A B M Shawkat Ali, Saleh A. Wasimi: Data Mining: Methods and Techniques, Cengage Learning.
8. Rajan Chattamvelli: Data Mining Methods, Narosa Publishing House
9. Sam Anahory, Dennis Murray: Data Warehousing in the real World, Pearson Education.
10. Pieter Adriaans, Dolf Zantinge: Data Mining, Pearson Education.

Web References:

1. <http://technet.microsoft.com/en-us/library/dd334421%28v=sql.100%29.aspx> [For Data Mining Analysis Services]
2. http://infogoal.com/datawarehousing/business_intelligence.htm [For architecture of Data Mining Applications]
3. <http://www.toadworld.com/platforms/sql-server/w/wiki/9478.introducing-data-mining-tutorial.aspx> [For Business Intelligence Tutorials and Videos]
4. <http://www.oracle.com/technetwork/database/bi-datawarehousing/dbbi-tech-info-094662.html> [For Video tutorials of data mining]

Course Outcomes:

- C01 : Describe what Data Science and Data Analytics are and the skill sets needed to be a data scientist.
- C02 : Understand significance of exploratory data analysis in statistical and visualization aspects.
- C03 : Understand and apply data analytics techniques such as supervised,unsupervised and EDA.

CO4 : Understand the importance of analytics in applications and able to implement them using data science techniques.

CO5: Able to understand the recent trends and future directions of data analytics.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction of Data Analytics	✓				
2	Exploratory Data Analysis and the Data Science Process		✓	✓		
3	Supervised Learning Methods			✓		
4	Unsupervised Learning Methods			✓		
5	Applied Data Science				✓	
6	Recent trends and future directions					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	3	1	-	2	3	1	1	1
CO2	2	2	2	3	1	1	2	2	1
CO3	1	1	2	2	1	2	1	3	3
CO4	3	1	-	1	-	-	3	-	2
CO5	1	1	3	-	3	1	-	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA924: Mini Project

(100 Marks)

Contact Hours: 06

Steps involved in the project work:

The complete project work should be done by the student only. The role of guide should be about guidance wherever any problem encounters during project. The following are the major steps involved in the project, which may help you to determine the milestones and regulate the scheduling of the project:

- Select a suitable topic of your interest.
- Prepare the project proposal in consultation with the project guide.
- Performa for Approval of Project duly filled and signed by both the student and Project Guide with date.
- Prepare project report according to the specific format.

Evaluation of project/dissertation work:

The project report shall normally be written in English in the specified format and shall be characterized by significant contribution to knowledge in the field. Normally two copies of the report are to be submitted for evaluation. The Project report prepared according to approved guidelines and duly signed by the supervisor(s) and the Head of the Department shall be submitted to the Head of the Institution. The evaluation scheme of Project is as under:

Course	Course Title	Teaching Scheme				Internal	End Semester Examination	Total
		Contact Hours			Credit	Continuous Evaluation	Viva	
		Inst.	Industry	Total				
CA924	Mini Project	6	-	6	6	30	70	100

The internal evaluation of project is done based on progress reports and internal presentations. The final evaluation of the project will be based on the project report submitted and a Viva-Voce Examination by a Board of Examiners.

If a candidate fails to submit the project report on or before the specified deadline, he/she is deemed to have failed in the Project Work and shall re-enroll for the same in a subsequent semester. If a candidate fails in the viva-voce examinations of Project work he/she shall resubmit the project report within specified duration decided by university. The resubmitted project will be evaluated during the subsequent academic session. A copy of the approved project report after the successful completion of viva examinations shall be kept in the library of the college / institution.

Course Outcomes :

CO1 : Student will understand the implementation of concepts of SDLC and Software Engineering.

CO2 :The programming concepts they learn during their academics, it will be converted in to the actual implementations.

CO3: Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.

CO4: Students must understand the difference between a program and professional application/product/software.

CO5: Students will learn different categories of applications like Desktop application, Web applications, etc.

Web References:

1. <http://www.shipoff.com/software-development-resources/>[Software development Resources]
2. <http://www.techrepublic.com/topic/software-development/>[Software development tips and tutorials]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	3	3	3	3	3	3	2	3
CO2	3	3	3	3	3	3	3	2	3
CO3	3	3	3	3	3	3	3	2	3
CO4	3	3	3	3	3	3	3	2	3
CO5	3	3	3	3	3	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

OBJECTIVES, TEACHING SCHEME & DETAILED SYLLABUS

FOR

**MCA Lateral PROGRAMME
(4th SEMESTER)**

**EFFECTIVE FROM
ACADEMIC YEAR 2018-19**

Semester-IV

Effective from Year 2018-19

Course Code	Course Title	Teaching Scheme				Internal	End Semester Examination		Total
		Contact Hours			Credit	Continuous Evaluation	Report	Presentation & Viva	
		Inst.	Industry	Total					
CA925	Project Work	2	28	30	30	200	200	400	800

CA925: Project Work

(800 Marks)

Contact Hours: 30

1. Project/dissertation work:

Students of MCA have to do the Project work in an Industrial/ Research Organization of computer field. Project work shall be carried out under the supervision of a qualified teacher in the Department as well as an expert from organization. Students have to meet the institute supervisor periodically and to attend the project/dissertation review meetings for evaluating the progress. The Project work shall be pursued for a minimum of 16 weeks during the semester.

1.1 MORI Principle to choose the topic:

Final semester project work is important and challenging element of MCA study. Selection of topic of project work is very crucial and important aspect of that. Following are important principles that will help students to select the appropriate topic of project work.

- **Manageable:** your project topic must be sufficiently focused so that it is possible for you to do the topic justice within the available time (one semester). You may have a real interest in, say, *'the impact of technology computers on Indian Economic Growth since Independence'*, but you certainly won't be able to cover this topic in any detail in the space of one semester
- **Original:** this relates to the above point, since a topic that is focused and manageable is more likely to be one that has not been written about too extensively, thus leaving room for your original contribution. Ideally you will find an interesting and well-chosen topic which will impress those marking your work.
- **Relevant:** your project should clearly be relevant to some aspect of your studies, but it might also be relevant to your plans for, say, postgraduate study or a career. The dissertation may also be relevant in the sense that it plays to some of your established strengths, such as a particular course module or topic that you have enjoyed studying and in which you have previously done well.
- **Interesting:** you are obviously more likely to enjoy and be successful in your dissertation if it is of real interest to you and to those marking your work. Ask yourself if you are sufficiently committed to your idea to be able to give it your best throughout the duration of your project. You should also ascertain whether your supervisor finds the idea interesting during your initial discussions with her or him.

1.2 Evaluation of project/dissertation work:

The project report shall normally be written in English in the specified format and shall be characterized by significant contribution to knowledge in the field. Normally two copies of the

report are to be submitted for evaluation. The Project report prepared according to approved guidelines and duly signed by the supervisor(s) and the Head of the Department shall be submitted to the Head of the Institution. The evaluation scheme of Project/Dissertation is as under:

Teaching Scheme			Credit	Internal	End Semester Examination		Total
Contact Hours				Continuous Evaluation	Report	Presentation & Viva	
Inst.	Industry	Total					
2	28	30	30	200	200	400	800

The internal evaluation of project is done based on progress reports and internal presentations.

The final evaluation of the project will be based on the project report submitted and a Viva-Voce Examination by a Board of Examiners.

If a candidate fails to submit the project report on or before the specified deadline, he/she is deemed to have failed in the Project Work and shall re-enroll for the same in a subsequent semester. If a candidate fails in the viva-voce examinations of Project he/she shall resubmit the project report within specified duration decided by university. The resubmitted project will be evaluated during the subsequent academic session. A copy of the approved project report after the successful completion of viva examinations shall be kept in the library of the college / institution.

Course Outcomes :

CO1 : Student will understand the implementation of concepts of SDLC and Software Engineering.

CO2 :The programming concepts they learn during their academics, it will be converted in to the actual implementations.

CO3: Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.

CO4: Students must understand the difference between a program and professional application/product/software.

CO5: Students will learn different categories of applications like Desktop application, Web applications, etc.

Web References:

1. <http://www.microtoolsinc.com/Howsrs.php>[For Software Requirement Specification Guidelines]
2. <http://techwhirl.com/writing-software-requirements-specifications/> [For goals of SRS]

3. <https://kepler-project.org/developers/reference/software-development-guidelines>[For Software Development Guidelines]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	3	3	3	3	3	3	2	3
CO2	3	3	3	3	3	3	3	2	3
CO3	3	3	3	3	3	3	3	2	3
CO4	3	3	3	3	3	3	3	2	3
CO5	3	3	3	3	3	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”